

Formal Verification Supplement: Quadratic Aggregation Certificates

1 Quadratic systems and the certificate question

Let $n, m \geq 1$, and let

$$f_i(x) = x^\top A_i x + 2b_i^\top x + c_i, \quad S = \{x \in \mathbb{R}^n : f_i(x) < 0 \text{ for } i = 1, \dots, m\}, \quad (1.1)$$

where $A_i \in \mathbb{S}^n$, $b_i \in \mathbb{R}^n$, and $c_i \in \mathbb{R}$. Here $\mathbb{S}^n$ denotes the real symmetric $n \times n$ matrices; $A \succeq 0$ and $A \succ 0$ mean positive semidefinite and positive definite, respectively. We use the Euclidean vector norm and the Frobenius matrix norm, and write $\langle A, B \rangle = \text{tr}(AB)$. The symbol $\text{conv } S$ denotes the ordinary convex hull, without taking its closure. A hull is *proper* if it differs from $\mathbb{R}^n$. Our question is whether a nonempty quadratic system has a proper hull, and how this can be certified by its defining inequalities.

For $\lambda \in \mathbb{R}^m$, write

$$(A_\lambda, b_\lambda, c_\lambda) = \sum_{i=1}^m \lambda_i (A_i, b_i, c_i), \quad f_\lambda = \sum_{i=1}^m \lambda_i f_i.$$

A nonzero $\lambda \geq 0$ defines an *aggregation*: $f_\lambda(x) < 0$ is valid for every $x \in S$. We call λ a *convex certificate* if $A_\lambda \succeq 0$, so that f_λ is convex on $\mathbb{R}^n$. It is *nontrivial* if $(A_\lambda, b_\lambda) \neq (0, 0)$. Otherwise the aggregated function is constant. If $S \neq \emptyset$, that constant is strictly negative, as evaluation at any point of S shows. We retain zero in the following cone for convenience:

$$\mathcal{K} = \{\lambda \in \mathbb{R}_+^m : A_\lambda \succeq 0\}, \quad \mathcal{T} = \{\lambda \in \mathbb{R}^m : A_\lambda = 0, b_\lambda = 0\}. \quad (1.2)$$

Thus a nontrivial convex certificate exists precisely when $\mathcal{K} \not\subseteq \mathcal{T}$.

The homogeneous matrices and map associated with (1.1) are

$$Q_i = \begin{pmatrix} A_i & b_i \\ b_i^\top & c_i \end{pmatrix}, \quad f_i^h(x, t) = x^\top A_i x + 2tb_i^\top x + c_i t^2, \quad f^h = (f_1^h, \dots, f_m^h). \quad (1.3)$$

We also write $Q_\lambda = \sum_i \lambda_i Q_i$ and $g_i(x) = x^\top A_i x$. Define the strict homogeneous feasible set $S^h = \{(x, t) : f_i^h(x, t) < 0 \text{ for all } i\}$. It is open and invariant under multiplication by every nonzero real scalar. For $t \neq 0$, $(x, t) \in S^h$ is equivalent to $x/t \in S$. For $\alpha \in \mathbb{R}^n \setminus \{0\}$ and $s \in \mathbb{R}$, put

$$H_{\alpha, s} = \{(x, t) \in \mathbb{R}^{n+1} : \alpha^\top x = st\}, \quad E = \{(x, t) : t = 0\}. \quad (1.4)$$

Definition 1.1 (Hidden hyperplane convexity). A homogeneous quadratic map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ has *hidden hyperplane convexity* (HHC) if $\phi(H)$ is convex for every linear hyperplane $H \subset \mathbb{R}^d$.

This is the definition of Blekherman, Dey, and Sun [3, Definition 2.1]; their work appeared as [4]. Ordinary *hidden convexity* means only that $\phi(\mathbb{R}^d)$ is convex. The two properties differ. All homogeneous quadratic images contain zero and are invariant under nonnegative scaling, so a convex such image is a convex cone; it need not be closed. HHC implies ordinary hidden convexity when $d \geq 3$: any two points of $\mathbb{R}^d$ lie in some linear hyperplane, whose image contains the segment joining their images.

Question 1.2 (Blekherman–Dey–Sun, Conjecture 3.3). Suppose $S \neq \emptyset$ and f^h has HHC. Is

$$\text{conv } S \neq \mathbb{R}^n \iff \exists \lambda \in \mathbb{R}_+^m : A_\lambda \succeq 0, \quad (A_\lambda, b_\lambda) \neq (0, 0)? \quad (1.5)$$

The formulation in Blekherman et al. [3, Conjecture 3.3] says that the aggregated function is not a negative constant. Under nonemptiness this is equivalent to the displayed condition. Their standing setting has $n \geq 3$ and $m \geq 2$; the formulation here also includes the smaller positive dimensions and numbers of inequalities. Throughout, numbered result locators for that work refer to the explicitly cited May 29, 2023 arXiv version; the earlier author manuscript has different numbering for several background results.

The reverse implication in (1.5) is elementary: a nonconstant convex quadratic aggregation bounds S by a proper convex sublevel set. The unresolved case in the cited conjecture concerns the opposite implication when $\mathcal{K}$ contains nonzero vectors but all of them belong to $\mathcal{T}$. Under convexity of $(g_1, \dots, g_m)(\mathbb{R}^n)$, Blekherman, Dey, and Sun already prove that $\mathcal{K} = \{0\}$ implies $\text{conv } S = \mathbb{R}^n$ [3, Proposition 2.14]. For $n \geq 2$, their separable case, with all Q_i diagonal, also already gives the certificate characterization [3, Theorem 2.12].

It is essential to distinguish this certificate question from a complete description of the hull. In the aggregation literature, a *good aggregation* is a nonnegative nonzero multiplier λ such that $\text{conv } S \subseteq \{x : f_\lambda(x) < 0\}$ and Q_λ has at most one negative eigenvalue. The function f_λ need not be globally convex. Blekherman, Dey, and Sun prove that these good aggregations describe $\text{conv } S$ under HHC, $n \geq 3$, and $\emptyset \neq S$ with proper hull [3, Theorem 2.9]. Thus an existence certificate and a family of hull-defining aggregations address different questions.

The two-constraint case belongs to established theory. Dines’ theorem gives convexity of the joint image of two homogeneous quadratic forms [6], and therefore HHC for a pair of homogeneous forms and their restrictions. Yildiran characterizes the hull of a nonempty set defined by two strict quadratic inequalities using at most two aggregations [7]. For three quadratic inequalities in dimension $n \geq 3$ with a nonempty proper hull, Dey, Muñoz, and Serrano obtain an aggregation description when the homogenized matrices Q_1, Q_2, Q_3 admit a positive definite real linear combination [5]. Here “positive definite linear combination” allows real coefficients of either sign; it does not mean a nonnegative aggregation. More recently, Blekherman and Dunbar use the spectral curve and algebraic topology to obtain further three-quadratic hull descriptions [2]. Their available preprint treats finiteness and related spectral conditions [1]. These results set the context for the certificate question, while neither the classical two-form convexity theorem nor an elementary separation lemma is itself a new contribution of the present work.

2 The certificate theorem

Only a family of hyperplanes approaching E is needed to answer Question 1.2. We first make this weaker assumption precise.

Definition 2.1 (Asymptotic hyperplane convexity). The map f^h has *asymptotic hyperplane convexity* (AHC) if, for every $\alpha \neq 0$ and every $R \in \mathbb{R}$, there exists $s > R$ such that $f^h(H_{\alpha,s})$ is convex.

Equivalently, for each $\alpha \neq 0$ there is a strictly increasing sequence (s_k) tending to $+\infty$ with all these images convex. Indeed, the sequence condition implies the unbounded-level condition; conversely, choose a good $s_1 > 1$ and recursively a good $s_{k+1} > \max\{s_k, k+1\}$. HHC implies AHC because $H_{\alpha,s}$ is a linear hyperplane for every s . This assumption concerns the images themselves and requires no closedness.

Theorem 2.2 (Convex aggregation certificate). *Suppose $S \neq \emptyset$ and f^h has AHC. Then*

$$\text{conv } S \neq \mathbb{R}^n \iff \exists \lambda \geq 0 : \quad A_\lambda \succeq 0, \quad (A_\lambda, b_\lambda) \neq (0, 0). \quad (2.1)$$

The certificate-to-proper-hull implication holds without either assumption.

Corollary 2.3 (Conjecture 3.3 of Blekherman–Dey–Sun). *If $S \neq \emptyset$ and f^h has HHC, then (2.1) holds.*

The corollary follows from HHC implying AHC. It answers the cited conjecture for every finite number of inequalities, including the case where nonzero convex certificates exist but all their quadratic and linear coefficients might vanish. The proof below shows that this case forces a whole-space hull. The theorem also isolates a weaker sufficient hypothesis; we attach no novelty claim to the separation or compactness facts it uses.

2.1 Hyperplanes beyond a supporting halfspace

Lemma 2.4 (No common strict negative quadratic direction). *If $\text{conv } S \neq \mathbb{R}^n$, then there is no $v \in \mathbb{R}^n$ satisfying $v^\top A_i v < 0$ for every i . Equivalently, $E \cap S^h = \emptyset$.*

Proof. If such a v existed, then for any fixed $y \in \mathbb{R}^n$,

$$f_i(y + rv) = r^2 v^\top A_i v + 2r(v^\top A_i y + b_i^\top v) + f_i(y) \longrightarrow -\infty \quad (|r| \longrightarrow \infty).$$

There are finitely many constraints. Hence both $y + rv$ and $y - rv$ belong to S for all sufficiently large positive r , and their midpoint y belongs to $\text{conv } S$. This holds for every y , a contradiction. Finally, $f_i^h(v, 0) = v^\top A_i v$ proves the equivalence. $\square$

This lemma concerns a simultaneous strict sign condition on the leading forms. It is not a characterization of the recession cone of S .

Lemma 2.5 (Sweeping hyperplanes). *Suppose $S \neq \emptyset$ and $\text{conv } S \neq \mathbb{R}^n$. There are $\alpha \neq 0$ and $\beta \in \mathbb{R}$ such that*

$$\alpha^\top x < \beta \quad (x \in S). \quad (2.2)$$

For any such α, β , one has $H_{\alpha,s} \cap S^h = \emptyset$ for every $s \geq \beta$.

Proof. The set S is open, and its convex hull is open as well: in a finite convex combination, one point with positive weight can vary in an open ball, giving an open ball around the combination. Separate a point $y \notin \text{conv } S$ from this nonempty open convex set. Finite-dimensional separation supplies $\alpha \neq 0$ with $\alpha^\top x \leq \alpha^\top y$ throughout $\text{conv } S$. Equality at a point of that open set would be contradicted by a small move in direction α . Thus (2.2) holds with $\beta = \alpha^\top y$.

If $(x, t) \in H_{\alpha,s} \cap S^h$ and $t = 0$, then Lemma 2.4 gives a contradiction. If $t \neq 0$, the identity $f_i^h(x, t) = t^2 f_i(x/t)$ gives $x/t \in S$, whereas $\alpha^\top(x/t) = s \geq \beta$, again a contradiction. This reasoning applies to either sign of t . $\square$

Lemma 2.6 (Separation on a hyperplane). *Suppose $s \neq 0$, the set $f^h(H_{\alpha,s})$ is convex, and $H_{\alpha,s} \cap S^h = \emptyset$. There exists $\lambda \geq 0$ with $\sum_i \lambda_i = 1$ such that, for every $x \in \mathbb{R}^n$,*

$$0 \leq x^\top A_\lambda x + \frac{2}{s}(\alpha^\top x)(b_\lambda^\top x) + \frac{c_\lambda}{s^2}(\alpha^\top x)^2. \quad (2.3)$$

Proof. The convex image $C = f^h(H_{\alpha,s})$ contains zero and is disjoint from the open negative orthant $N = -\mathbb{R}_{++}^m$. Separation of these two convex sets gives a nonzero λ and a real a with $\lambda^\top z \leq a \leq \lambda^\top y$ for $z \in N$, $y \in C$. If some $\lambda_i < 0$, sending z_i to $-\infty$ with the other coordinates fixed contradicts the upper bound. Thus $\lambda \geq 0$, and $\sup_{z \in N} \lambda^\top z = 0$. Since $0 \in C$, necessarily $a = 0$, and $\lambda^\top f^h \geq 0$ on $H_{\alpha,s}$. Divide λ by its positive coordinate sum. Substituting $(x, t) = (x, \alpha^\top x/s)$ gives (2.3). Closedness of C was not used. $\square$

2.2 A nonzero limit in coefficient space

The relevant cone retains the quadratic and linear coefficients together:

$$P = \text{cone}\{(A_i, b_i) : 1 \leq i \leq m\} = \{(A_\lambda, b_\lambda) : \lambda \geq 0\} \subseteq \mathbb{S}^n \times \mathbb{R}^n. \quad (2.4)$$

We equip this space with $\|(A, b)\| = (\|A\|_F^2 + \|b\|^2)^{1/2}$. Closedness of this particular cone is essential: a limit in its closure must correspond to actual aggregation weights. The following elementary fact supplies that property.

Lemma 2.7 (Finite coefficient cones are closed). *The cone generated by a finite family in a finite-dimensional real vector space is closed.*

Proof. Every conic combination can be represented using a linearly independent subfamily. To see this, choose a representation with the fewest positive coefficients. If its active generators are dependent, take a nonzero relation $\sum_i d_i v_i = 0$, reversing its sign so some $d_i > 0$. For the active coefficients $a_i > 0$, set $\tau = \min_{d_i > 0} a_i/d_i$. Then all $a_i - \tau d_i$ are nonnegative, at least one vanishes, and the represented point is unchanged, a contradiction. For a linearly independent subfamily, the coefficient map is a linear homeomorphism onto its span; that span is closed, so the image of the nonnegative orthant is closed. There are only finitely many subfamilies. The original cone is their finite union and is closed. The empty subfamily accounts for the zero vector. $\square$

Proof of Theorem 2.2. First suppose a nontrivial convex certificate λ exists. Its nonconstant coefficient pair implies $\lambda \neq 0$, so $f_\lambda < 0$ on S . Positive semidefiniteness makes f_λ convex, whence $\text{conv } S \subseteq \{f_\lambda < 0\}$. This sublevel set is proper. Indeed, if $A_\lambda \neq 0$, it has a unit eigenvector v with eigenvalue $\mu > 0$, and $f_\lambda(rv) = \mu r^2 + 2rb_\lambda^\top v + c_\lambda \rightarrow +\infty$ as $r \rightarrow +\infty$. If $A_\lambda = 0$, then $b_\lambda \neq 0$ and $f_\lambda(rb_\lambda) = 2r\|b_\lambda\|^2 + c_\lambda \rightarrow +\infty$. This proves the easy implication without using nonemptiness or AHC.

For the converse, suppose $S \neq \emptyset$, f^h has AHC, and $\text{conv } S \neq \mathbb{R}^n$. Fix $x_0 \in S$, and choose α, β from Lemma 2.5. AHC gives positive levels $s_k \rightarrow +\infty$ with $s_k \geq \beta$ and convex images $f^h(H_{\alpha,s_k})$. Lemmas 2.5 and 2.6 give nonzero normalized weights $\lambda^k \geq 0$ satisfying (2.3). Write $(A_k, b_k, c_k) = (A_{\lambda^k}, b_{\lambda^k}, c_{\lambda^k})$. Strict feasibility gives the crucial upper bound

$$c_k < -x_0^\top A_k x_0 - 2b_k^\top x_0 =: L(A_k, b_k), \quad (2.5)$$

where L is a fixed continuous linear functional on coefficient space. The multiplier of c_k in (2.3) is a square. Replacing c_k by its upper bound therefore yields

$$0 \leq x^\top A_k x + \frac{2}{s_k}(\alpha^\top x)(b_k^\top x) + \frac{L(A_k, b_k)}{s_k^2}(\alpha^\top x)^2 \quad (x \in \mathbb{R}^n). \quad (2.6)$$

Moreover, $(A_k, b_k) \neq (0, 0)$: otherwise (2.5) gives $c_k < 0$, and the original inequality (2.3) at $x = \alpha$ reads $0 \leq c_k \|\alpha\|^4 / s_k^2 < 0$.

Set $\rho_k = \|(A_k, b_k)\| > 0$ and $(\hat{A}_k, \hat{b}_k) = (A_k, b_k) / \rho_k$. These pairs lie in the unit section of P , which is compact by Lemma 2.7. Passing to a subsequence, let

$$(\hat{A}_k, \hat{b}_k) \longrightarrow (A_*, b_*) \in P, \quad \|(A_*, b_*)\| = 1.$$

Divide (2.6) by ρ_k . By linearity of L , its right-hand side becomes

$$x^\top \hat{A}_k x + \frac{2}{s_k} (\alpha^\top x) (\hat{b}_k^\top x) + \frac{L(\hat{A}_k, \hat{b}_k)}{s_k^2} (\alpha^\top x)^2.$$

For every fixed x the last two terms tend to zero. Consequently $x^\top A_* x \geq 0$ for every x . This conclusion uses one coefficient subsequence for all x ; no point-dependent subsequence is chosen. The symmetric matrix A_* is positive semidefinite. Since $(A_*, b_*) \in P$, there are actual weights $\lambda^* \geq 0$ with $(A_{\lambda^*}, b_{\lambda^*}) = (A_*, b_*)$. Their coefficient pair has norm one, so they give the required nontrivial convex certificate. $\square$

Remark 2.8 (What the limit proves). Normalizing the weights in the simplex alone can give a negative constant aggregation in the limit. Normalizing the nonconstant coefficient pair prevents this loss of information. Bound (2.5) removes the constant before normalization; no bound on c_k / ρ_k is assumed. The final weights need not be a limit of the original weights, or of their rescalings. Their existence follows from closedness of the finitely generated cone P .

Remark 2.9 (Convexity along one normal). The hard implication needs convexity of $f^h(H_{\alpha,s})$ at arbitrarily large levels for only one α satisfying (2.2). AHC is a convenient condition that can be stated without knowing a supporting normal. For fixed α , the hyperplanes $H_{\alpha,s}$ approach E as $s \rightarrow +\infty$; convexity of $f^h(E)$ alone is not the hypothesis used in the proof. There is no restriction beyond $n, m \geq 1$. For one constraint, a negative quadratic direction gives a whole-space hull by Lemma 2.4; the remaining cases follow from the easy implication or a negative constant. For two constraints, Dines' theorem makes HHC automatic, recovering the known certificate case discussed in Section 1.

2.3 A quantitative form of the obstruction

For completeness, we give a second way to exclude degeneration to constant certificates. It supplies a uniform bound on the sweep level under the hypothesis that all convex certificates are trivial. The cone estimate itself is a standard compactness argument.

Lemma 2.10 (Uniform cone separation). *Let P, D be closed cones containing zero in a finite-dimensional normed space, with $P \cap D = \{0\}$. Then some $\kappa > 0$ satisfies*

$$\text{dist}(p, D) \geq \kappa \|p\| \quad (p \in P).$$

Neither cone is required to be convex.

Proof. If $P = \{0\}$, any positive κ works. Otherwise the continuous function $p \mapsto \text{dist}(p, D)$ has a positive minimum κ on the compact set $P \cap \{\|p\| = 1\}$: a zero distance would put a unit point of P in the closed set D . Scaling both cones proves the bound for $p \neq 0$, and the assertion at zero follows from $0 \in D$. $\square$

Apply the lemma to P in (2.4) and $D = \mathbb{S}_+^n \times \mathbb{R}^n$. The condition $P \cap D = \{0\}$ is exactly the assertion that every convex certificate is trivial. If A_- is the positive semidefinite negative part of a symmetric matrix A , then

$$\text{dist}((A, b), D) = \|A_-\|_F \leq \sqrt{n} \max\{0, -\lambda_{\min}(A)\}. \quad (2.7)$$

For the equality, diagonalize A orthogonally. In that basis every positive semidefinite approximant has nonnegative diagonal entries; the sum of squared errors is at least the sum of squares of the negative eigenvalues of A . Replacing those eigenvalues by zero achieves this bound, and the b component can be left unchanged. The inequality follows by bounding each negative eigenvalue in absolute value by $\max\{0, -\lambda_{\min}(A)\}$.

Proposition 2.11 (Bound on a sweep certificate). *Suppose every convex certificate is trivial, and let κ be as in Lemma 2.10 for the preceding cones. If $\lambda \geq 0$, $c_\lambda < 0$, $s > 0$, $\alpha \neq 0$, and (2.3) holds, then*

$$s \leq \frac{2\sqrt{n}\|\alpha\|}{\kappa}. \quad (2.8)$$

Proof. The inequality at $x = \alpha$ shows that $\rho = \|(A_\lambda, b_\lambda)\|$ cannot vanish. The cone estimate and (2.7) give $\lambda_{\min}(A_\lambda) \leq -\kappa\rho/\sqrt{n}$. Choose a unit eigenvector u for this eigenvalue. Evaluating (2.3) at u , discarding the nonpositive constant term, and using $\|b_\lambda\| \leq \rho$ gives

$$0 \leq -\frac{\kappa}{\sqrt{n}}\rho + \frac{2\|\alpha\|}{s}\rho.$$

Division by $\rho > 0$ proves (2.8). □

This gives an alternative completion of the hard implication. Under its hypotheses, suppose every convex certificate is trivial and choose the normalized sweep weights λ^k as before. Compactness of the simplex gives a subsequence converging to $\bar{\lambda}$, with $\bar{\lambda} \geq 0$ and $\sum_i \bar{\lambda}_i = 1$. The coefficients b_k, c_k are bounded, so letting $k \rightarrow \infty$ in (2.3) shows $A_{\bar{\lambda}} \succeq 0$. Triviality then gives $A_{\bar{\lambda}} = 0$, $b_{\bar{\lambda}} = 0$, and $c_{\bar{\lambda}} < 0$ by strict feasibility. Thus $c_k < 0$ eventually along this subsequence. Proposition 2.11 contradicts $s_k \rightarrow +\infty$. The constant κ depends only on the original quadratic and linear coefficients; this argument neither bounds aggregation weights nor provides a procedure for verifying AHC.

3 Formal verification of the certificate theorem

A Lean development verifies Theorem 2.2 and its HHC specialization, Corollary 2.3, for finite families of real symmetric matrices, real vectors, and real constants. Its strict feasible set, ordinary convex hull, and nontrivial certificate agree with those defined here. The main declaration is `proper_hull_iff_certificate` in the namespace `QuadraticAggregation.System`; the HHC specialization is `proper_hull_iff_certificate_of_hhc`. The assumptions are nonemptiness and AHC, or nonemptiness and HHC, respectively. No positive definite combination is assumed. The certificate-to-proper-hull implication is proved separately without either assumption.

The formal definitions express HHC through kernels of nonzero linear functionals, which are precisely the linear hyperplanes in this finite-dimensional setting. AHC is stated using arbitrarily large good levels, as in Definition 2.1; its equivalence with the increasing sequence formulation is also proved. Matrix positive semidefiniteness is linked to nonnegativity of the quadratic form, using the symmetry of the original matrices and their aggregations.

The proof follows the constant-elimination and coefficient-normalization argument in Section 2. In particular, it constructs separation weights from the convex hyperplane images and obtains the

nonzero limit in the actual finitely generated coefficient cone. These are proved steps, not additional hypotheses of the headline theorem. The implementation uses the entrywise maximum matrix norm and the product maximum norm for coefficient pairs. These are equivalent to the Frobenius and Euclidean norms in finite dimension; the compactness proof works in either choice.

The verified supporting results include Lemmas 2.4, 2.5, 2.6, and 2.7, and the uniform cone-separation lemma, Lemma 2.10. The last lemma is checked independently, although the main formal proof does not use it. The spectral distance calculation and quantitative alternative in Section 2.3 are not formalized as separate results.

This core package consists of eleven modules in `supplement/lean/Formal/QuadraticAggregation/`, accompanied by a declaration coverage map and reproducible checks. With the pinned Lean 4.33.1 toolchain, targeted compilation without warnings, a transitive axiom audit of all 178 declarations owned by those modules, and kernel replay of each module passed. The checks were rerun during preparation of this manuscript, with source fingerprints unchanged throughout the run. The only permitted axioms were `propext`, `Classical.choice`, and `Quot.sound`. Kernel replay uses the pinned Lean kernel and imported dependencies; it does not constitute verification by an independently implemented proof assistant or a replay of all dependencies from source.

This formal scope concerns existence of a nontrivial globally convex aggregation. It does not establish that such aggregations describe the entire hull. The closed-system and relaxation consequences, SDP characterizations, examples, applications, further aggregation results, and literature-priority claims are outside this core package. Separate packages verify the specified later results described in the accompanying formal accounts. Formal verification of the stated theorem does not certify the rest of the manuscript.

References

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